Virtual Model Control of a Soft Continuum Manipulator

Paper brief and pre-submission checklist

1 Purpose of this document

This is a working brief for organizing the repository and finishing the remaining experiments before the paper is submitted. It is meant to be read by a coding assistant with access to the actual repository, so it states what exists, what is missing, and what should not be decided without asking. It does not target a specific venue; that is irrelevant to the work described here.

2 Project summary

The paper applies Virtual Model Control (VMC) to a tendon-driven soft continuum manipulator (three segments, Piecewise Constant Curvature). VMC attaches virtual mechanical elements (springs, dampers, and nonlinear variants) between points on the robot and reference points, and maps the resulting virtual forces to actuator torques through the Jacobian. The controller needs only a forward-kinematic model, not system dynamics, and remains passive by construction.

Four contributions frame the paper:

1. A framework for whole-body control that plans in both pose and impedance space, not just pose.
2. The first application of VMC to a soft continuum arm, with one simple control framework covering tip and full-body pose and stiffness.
3. Intuitive definition and placement of virtual elements (the main practical advantage over classical impedance control).
4. Native handling of obstacles, spanning avoidance and deliberate contact exploitation, not just avoidance.

People and repository

- Implementation: Ittirath Wiwatwanont (MSc Robotics, semester project).
- Supervision: Lorenzo Vignoli, day-to-day supervisor; Prof. Josie Hughes, PI.
- Repository: `https://github.com/vigno0405/SoftVMC` (referenced in the semester report; not accessible from this environment, so this brief describes the intended structure rather than the actual current file layout, work from the real local checkout).

3 Technical background

3.1 Kinematic model

The manipulator is described with a Piecewise Constant Curvature (PCC) approximation: each of the three segments is a circular arc, parametrized in a singularity-free representation to avoid the discontinuities of the standard bending-angle parametrization near the straight configuration. Virtual elements can attach anywhere along the body, not only the tip, using a normalized arc-length coordinate

per segment. PCC is exact only under pure bending moments and degrades under large external loads, this is a known modeling limitation, not a bug.

3.2 VMC formulation

Virtual elements produce a force from the displacement of a robot point relative to a reference:

$$\Delta x_i = x_{\text{ref},i} - x_i, \qquad f_i = k_i \Delta x_i - c_i \Delta \dot{x}_i,$$

mapped to joint torques via the principle of virtual work,

$$\tau = \sum_i J_i^T f_i.$$

Four virtual element families are already implemented:

- **Linear spring/damper** – the base element, $f = k\Delta x - c\Delta \dot{x}$.
- **Tanh (force-limited) spring** – caps the virtual force at a prescribed $F_{\max}$.
- **Constrained ("cart") linear spring** – projects the force onto a chosen axis, confining motion to that direction.
- **Gaussian repulsive field** – a local, distance-decaying repulsive force used to emulate obstacles, static or moving.

3.3 Software architecture

ROS 2 node graph: a GUI or custom Experiment Script publishes reference positions and virtual-element parameters; the VMC Torque Node (wrapping an internal "VMC Multi Modular" class) computes torque and virtual force from joint positions/velocities and load-cell shear force; the Dynamixel Driver actuates the robot. Killing the VMC Torque Node saves all published/subscribed values as `.npz` and `.png` for later analysis. A VMC Configuration file sets the number and type of springs/dampers for a given run. A GUI exists for quick tuning; scripts are preferred for repeatable experiments. A load-cell setup (string from end-effector to load cell) is used whenever shear force needs to be measured.

3.4 Known limitations (to carry into the Discussion section)

- **Force-transmission gap.** The virtual force is a control set-point, not a calibrated prediction: for the tanh spring, measured shear force saturates at $\approx$0.72 N against a 4 N virtual set-point, and the virtual-to-measured ratio drifts from $\approx$8 at small displacement to $\approx$5.5 near saturation, not a fixed scale factor. A friction-based transmission model has been proposed to explain this:

  $$f_u \cdot (J_u)^{-1} \cdot J_d^T \cdot f_d \cdot \mu, \qquad \mu \in \{0.4,\ 0.5\} \text{ (candidate values to fit)},$$

  where f_u is the measured force, f_d the commanded virtual force, and J_u, J_d the relevant Jacobians. This needs to be fit against the existing tanh-sweep data (Fig. 10 in the semester report) and validated, not asserted.
- **Encoder insensitivity.** Motor encoders barely register externally induced deformation (tendons resist external displacement while current flows), so the encoder-based configuration estimate can diverge from the true shape during contact. Demonstrated in the surface-tracking experiment ($y_{\max} = 0.003$ m inferred vs. a visibly larger real diagonal excursion).
- **Push/pull tendon asymmetry.** PCC assumes each segment can be both pulled and pushed; tendons can only pull, so over-release produces slack that the encoder does not detect, corrupting the state estimate. Mitigated with an angle-dependent pretension heuristic:

  $$\tau_{\text{applied}} = \tau_{\text{cmd}} - \gamma \max\big(0,\ \theta_{\text{rel}} - \theta_0\big), \qquad \theta_0 = 30^\circ,$$

  a partial fix, not a resolution; true fix is direct cable sensing (future work).

4 Paper structure

A. Introduction

Fig. 1 (concept/graphical abstract) plus a related-work comparison table (not a figure). The comparison table should contrast this work against prior VMC applications, which are all on rigid platforms:

- Pratt et al., *Virtual model control: an intuitive approach for bipedal locomotion*, IJRR 2001 – rigid bipedal locomotion, linear elements only.
- Zhang, Larby, Iida, Forni, *Virtual model control for compliant reaching under uncertainties*, IROS 2024 – rigid 8-DoF force-controlled arm, reaching around obstacles. Closest prior VMC work; worth naming precisely since Forni is already a collaborator on this project.
- Della Santina, Duriez, Rus, *Model-based control of soft robots: a survey*, IEEE Control Systems Magazine 2023 – soft robots, but model-based (needs dynamics), not VMC.

This work: soft continuum arm, PCC kinematics only (no dynamics identification), four virtual element families, avoidance and contact exploitation in one framework.

B. Method

- Fig. 2 – virtual elements attached to the body, plus a panel on the properties of those elements (how each family shapes force vs. displacement).
- Fig. 3 – experimental setup: robot and load-cell configuration.
- Fig. 4 – stiffness characterization: force saturation and forced-oscillation (torque disturbance) response. This lives in Methods, not Results, it establishes that the base linear element behaves as intended before the paper moves on to the other element types.

C. Results

C.a – Virtual elements for shaping of tip and body

- Fig. 5 – tanh element: virtual vs. measured shear, tanh vs. linear force profiles. This is where the friction-transmission model (Section 3.4) is fit and reported.
- Fig. 6 – constrained (cart) element.
- Fig. 7 – Gaussian element, combined with stiffness k, for reaching and avoidance planning together (see open question in Section 6).

C.b – Contact-based interaction and obstacle avoidance

- Fig. 8 – impedance shaping with contact and whole-body behavior, one four-panel figure: yarn-ball contact, compliant surface tracking, linear spring with/without a UR5 obstacle, and one vs. three distributed springs.
- Fig. 9 – exploiting load-cell contact for button pressing (new experiment, see Section 6).

C.c – Demo

- Fig. 10 – demonstration involving a person, framed around a possible medical/assistive use case: avoiding a person vs. touching them lightly while completing a task. Still undecided in scope (see Section 6).

D. Discussion and conclusion

The three known limitations from Section 3.4, with the force-transmission gap now partly addressed by the fitted friction model rather than left as an open list of possible causes.

5 Figure manifest: old report numbering to new paper numbering

"Old" refers to the figure numbers in the semester report (*SoftArmVMCSemester.pdf*). Everything marked *ready* reuses an existing figure or dataset from that report; everything marked *new* does not exist yet.

New	Content	Old source	Status
1	Concept / graphical abstract	none	new (design)
2	Virtual element family + properties	none directly; GUI (old Fig. 7) may inform it	new (design)
3	Setup: ROS graph + load cell	old Fig. 1, old Fig. 8	ready
4	Stiffness char.: saturation + disturbance	old Fig. 2a, 2b	ready
5	Tanh element + friction fit	old Fig. 3b, 10a, 10b	ready + new analysis
6	Constrained (cart) element	old Fig. 11	ready
7	Gaussian element + k combination	old Fig. 6a, 6b, 12	ready, scope open (§6)
8	Contact/shaping, 4 panels	old Fig. 3a, 4, 5a, 5b	ready
9	Button-pressing exploit/avoid	none	new (to run)
10	Demo with a person	none	new, undecided (§6)

Two old figures currently have no slot in the new plan and should not be silently dropped or silently reinserted:

- Old Fig. 9 (energy consumed during the stiffness sweep, Appendix C.1) – not listed under new Fig. 4 or anywhere else. Confirm whether it becomes a third panel in Fig. 4, moves to supplementary material, or is cut.
- Old Fig. 7 (GUI screenshot, Appendix A) – not listed anywhere in the new plan. Assume dropped from the print figures unless told otherwise; keep it available for the repository README.

6 New experiments: specification

6.1 Fig. 9 – button-pressing, avoid / exploit / minimize contact force

Reuses the existing load-cell setup and VMC configuration-file pattern, no new hardware beyond a button or button-like target instrumented so contact force is read through the load cell already in the loop.

- **Conditions to compare** (same target, same approach direction, different virtual-element configuration):
  1. *Avoid* – a repulsive (Gaussian) element keeps the tip clear of the target; task is not completed, this is the baseline showing pure avoidance.
  2. *Exploit* – an attractive element drives firm, sustained contact against the target.
  3. *Minimize* – a force-limited (tanh) or closed-loop element presses just enough to register contact and no more.

- **Metric** – peak and steady-state contact force from the load cell per condition, plus whether the task (button registers) succeeds, across repeated trials per condition.
- **Reuses** – the same Experiment-Script/config-file/logging pattern as the existing experiments, so the new runs drop into the same analysis pipeline without a new one being written from scratch.

This is the strongest new result for demonstrating contact exploitation: it is concrete, quantifiable off existing hardware, and directly supports contribution 4 in Section 2.

6.2 Fig. 7 – open question on scope

"Combination with k, reaching and avoidance planning" could mean two different things, and they imply different amounts of new work:

- **(a) Reuse.** Present the existing static (old Fig. 6a) and dynamic (old Fig. 12, $k_{\text{lin}} = 5$ vs. $50\,\text{N/m}$) Gaussian results together as one figure. No new data.
- **(b) New sweep.** Run a finer sweep across several k values (not just two points) to show a continuous trend between avoidance-dominated and reach-dominated behavior. New data collection.

Do not assume (b) without confirming, it changes the remaining experimental workload materially.

6.3 Fig. 10 – open question on protocol

Still marked undecided by the supervisors. Two practical points to resolve before scheduling it, not before writing about it:

- What the task actually is (a specific reach-while-avoiding or reach-while-touching task, ideally one that mirrors a concrete assistive/medical scenario rather than a generic demo).
- A person's image or video will appear in the paper; get informed consent from whoever is filmed before treating any footage as camera-ready material.

7 Repository organization checklist

For whoever (human or agent) has access to the actual local repository:

1. **Inventory.** List every experiment script, VMC configuration file, and logged output currently in the repo. Match each to the old semester-report figure it produced (2a, 2b, 3a, 3b, 4, 5a, 5b, 6a, 6b, 9, 10, 11, 12) using filenames, timestamps, and config parameters.
2. **Verify the log format.** The semester report states that killing the VMC Torque Node saves `.npz` and `.png` files, but this has not been checked against the actual repository. Search the codebase for where logging happens (e.g. `np.savez`, `np.save`, or any node-shutdown/cleanup handler) and for existing `.npz` files on disk. Report, per experiment, whether raw `.npz` data actually exists, only `.png` plots exist, or neither, since figures cannot be regenerated or restyled without the underlying `.npz`.
3. **Re-map.** Using the manifest in Section 5, organize outputs (copies or symlinks, not the raw logs themselves) into a structure that mirrors the paper, e.g. `paper/figures/fig04_stiffness/`, `paper/figures/fig08_contact_shaping/`, one folder per new figure number.
4. **Scaffold the new experiments.** For Fig. 9, and for Fig. 7 if scope (b) is confirmed, create new experiment scripts and VMC configuration files following the existing pattern (same node graph, same GUI/script choice, same `.npz`/`.png` logging), so new runs are consistent with the old ones and drop into the same analysis/plotting pipeline.
5. **Plotting consistency.** Reuse the color conventions and plotting style already established per old figure (e.g. the stiffness colorbar, the red/blue linear-vs-alternative convention) so the new and old figures look like they belong to the same paper.

6. **Provenance.** For every figure, keep a one-line note of which raw run(s), file, date, config, produced it, so it can be regenerated without repeating the physical experiment.
7. **Don't guess.** If a script or dataset matching an old figure cannot be found, or if either open question in Section 6 has not been resolved, surface it rather than deciding it unilaterally.

8 Reference list (for the comparison table / bibliography)

1. C. Della Santina, C. Duriez, D. Rus, "Model-based control of soft robots: a survey of the state of the art and open challenges," *IEEE Control Systems Magazine*, 43(3), 2023.
2. J. Pratt, C.-M. Chew, A. Torres, P. Dilworth, G. Pratt, "Virtual model control: an intuitive approach for bipedal locomotion," *IJRR*, 20(2), 2001.
3. Y. Zhang, D. Larby, F. Iida, F. Forni, "Virtual model control for compliant reaching under uncertainties," IROS 2024.
4. C. Della Santina, A. Bicchi, D. Rus, "On an improved state parametrization for soft robots with piecewise constant curvature and its use in model based control," *IEEE RA-L*, 5(2), 2020.
5. F. Stella, Q. Guan, C. Della Santina, J. Hughes, "Piecewise affine curvature model: a reduced-order model for soft robot-environment interaction beyond PCC," RoboSoft 2023.